

A Project Report Presentation:

# Real-Time Telemetry Analytics & Predictive Modelling

for Motorsports Strategy Optimization

SUBMITTED TO:

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# INTRODUCTION

## The F1 Data Challenge

**500 GB+**

telemetry data per race weekend

**300+**

sensors per car

**\$500K+**

cost of proprietary systems

## Evolution of Telemetry



**1980s**

Analog radio: RPM & fuel



**1990s**

Digital + GPS; McLaren Atlas



**2000s**

Real-time wireless; 2 Mbps



**2010s**

Big data ML, cloud computing



**2020s**

AI-driven, edge inference

# PROBLEM STATEMENT

## Financial Barriers

- McLaren ATLAS: \$200K+
- Software licenses: \$550K/yr
- Personnel: \$580K/yr

## Vendor Lock-in

- Closed-source systems
- No ML integration
- Propriety file formats

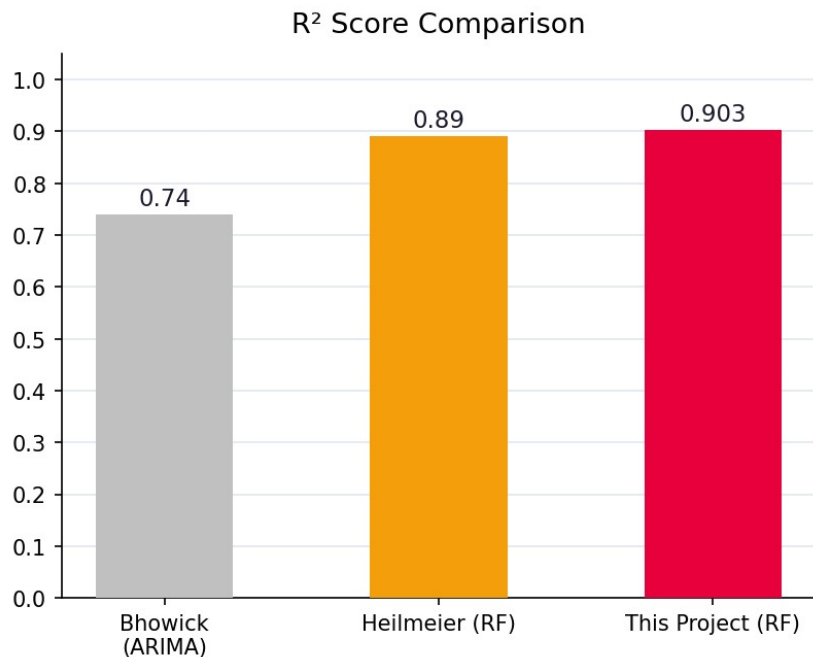
## Storage Inefficiency

- Full-rate capture = TBs
- 36,000 pts per session
- No intelligent sampling

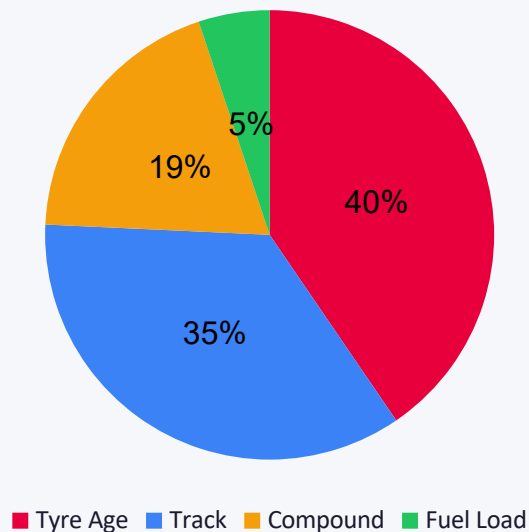
## Prediction Gap

- Rule-based tools only
- No real-time ML inference
- Post-race analysis only

# LITERATURE REVIEW



Feature Importance (Random Forest)



*Key insight: RF outperforms ARIMA/LR; Tyre Age is the dominant predictor at 40.46%*

# SYSTEM ARCHITECTURE

## Data Source Layer

F1 2018 Game (UDP · 60Hz · 1289B) / FastF1 API (Official FIA data)



## Data Layer

MySQL (sessions · laps · telemetry · strategy\_events) / File system



## Application Layer

Flask REST API · capture\_telemetry · analyze\_performance · ml\_lap\_prediction



## Presentation Layer

Web UI (Chart.js Dashboard) · CLI Tools · Batch Scripts

# END-TO-END DATA PIPELINE

## Data Ingestion

- UDP socket
- Port 20777
- 60 pkt/s

## Binary Parsing

- struct.unpack
- 1289-byte packet
- Validation 60s–180s

## Transformation

- Lap detection (timer delta)
- Sampling 60:1 → 1 Hz

## Storage

- MySQL INSERT →
- sessions
- laps
- telemetry

## Analysis & ML

- Statistics
- Charts
- Random Forest Training

## Presentation

- Flask API
- Chart.js Dashboard
- Strategy Advisor

# MACHINE LEARNING PIPELINE

1

## Data Extraction

SQL: SELECT lap\_time, tyre\_age, fuel\_load, compound, track WHERE is\_valid=1

2

## Feature Engineering

One-hot encode compound (6) + track (9) = 17 features total

3

## Train-Test Split

80% train (698 laps) · 20% test (175 laps) · random\_state=42

4

## Model Training

Random Forest (100 trees, n\_jobs=-1) vs Linear Regression baseline

5

## Evaluation

$R^2=0.91$  · MAE=1.4s · RMSE=2.3s → RF selected

6

## Serialization

joblib → best\_model.pkl + feature\_names.pkl (487 KB)

## Model Comparison

| Metric      | Linear Reg. | Random Forest |
|-------------|-------------|---------------|
| $R^2$ Score | 0.35        | 0.91 ✓        |
| MAE         | 0.93 s      | 1.40 s        |
| RMSE        | 4.12 s      | 2.30 s        |
| Train Time  | 2 s         | 45 s          |
| Non-linear  | ✗           | ✓             |

# DATABASE SCHEMA (3NF)

## sessions

**PK session\_id**

track\_name

session\_type

weather

date

## strategy\_events

**PK event\_id**

**FK lap\_id**

event\_type (ENUM)

duration\_sec

## laps

**PK lap\_id**

**FK session\_id**

lap\_number

lap\_time\_ms (INT)

tyre\_compound (ENUM)

tyre\_age

fuel\_load

is\_valid

## telemetry

**PK telemetry\_id**

**FK lap\_id**

speed

throttle

brake

gear

RPM

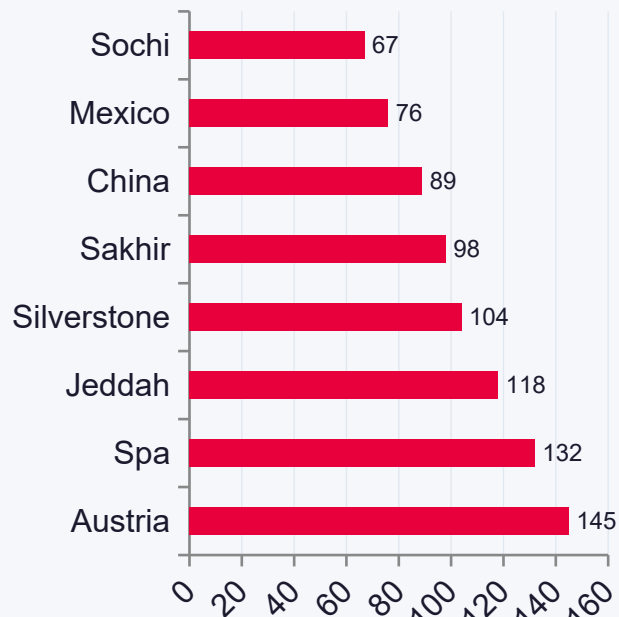
DRS

# DATASET & VALIDATION

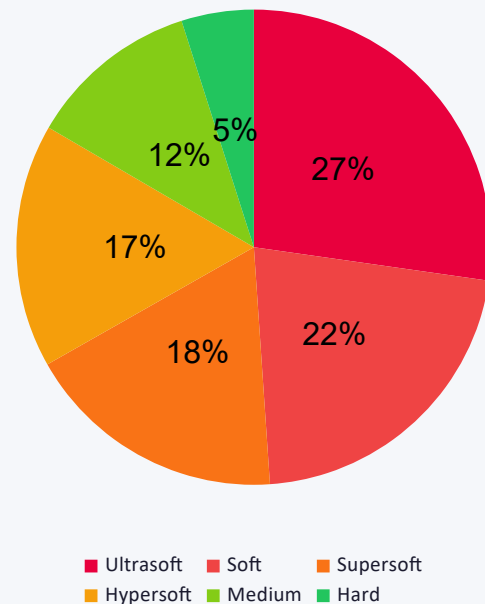
## Dataset Stats

|                |           |
|----------------|-----------|
| Total Laps     | 912       |
| Valid Laps     | 873       |
| Unique Tracks  | 9         |
| Tyre Compounds | 6         |
| Sessions       | 15        |
| Avg Lap Time   | 87.34 s   |
| Train Split    | 698 (80%) |
| Test Split     | 175 (20%) |

## Laps by Track



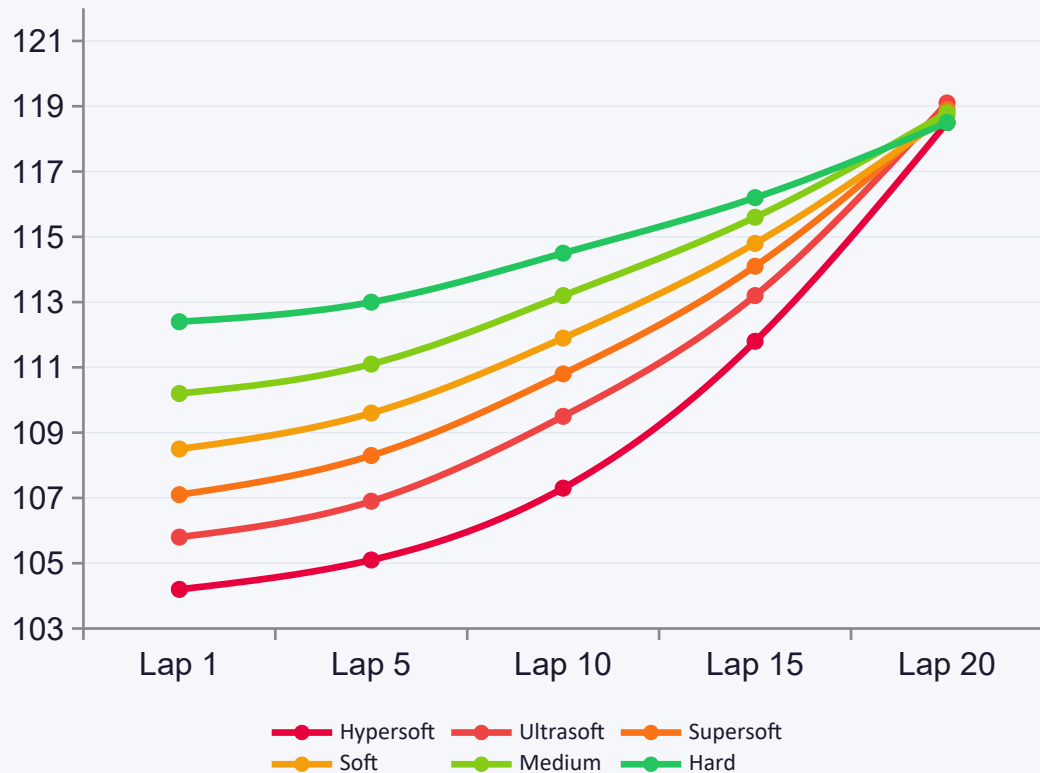
## Tyre Distribution



Source: F1 2018 Game (UDP capture) + FastF1 API (validation) · 5-fold CV MAE = 1.48s ± 0.06s

# RESULTS: TYRE DEGRADATION CURVES

Lap Time vs Tyre Age (seconds)



## Key Insights

- 1 Softer compounds: faster start, faster degradation
- 2 All compounds converge ~118-119s by lap 20
- 3 Hypersoft: 2.2s/lap faster initially, +14.3s degradation
- 4 Hard tyres: slowest start, best longevity (+6.1s total)
- 5 Tyre cliff at laps 15-20 — optimal pit window
- 6 Pit stop time ~23s → strategy depends on cliff timing

# EMPIRICAL EVALUATION

**0.908**

**R<sup>2</sup> Score**

Target  $\geq 0.85$  — Exceeded

**1.40s**

**MAE**

Target  $\leq 2.0s$  — 30% better

**47ms**

**Prediction Latency**

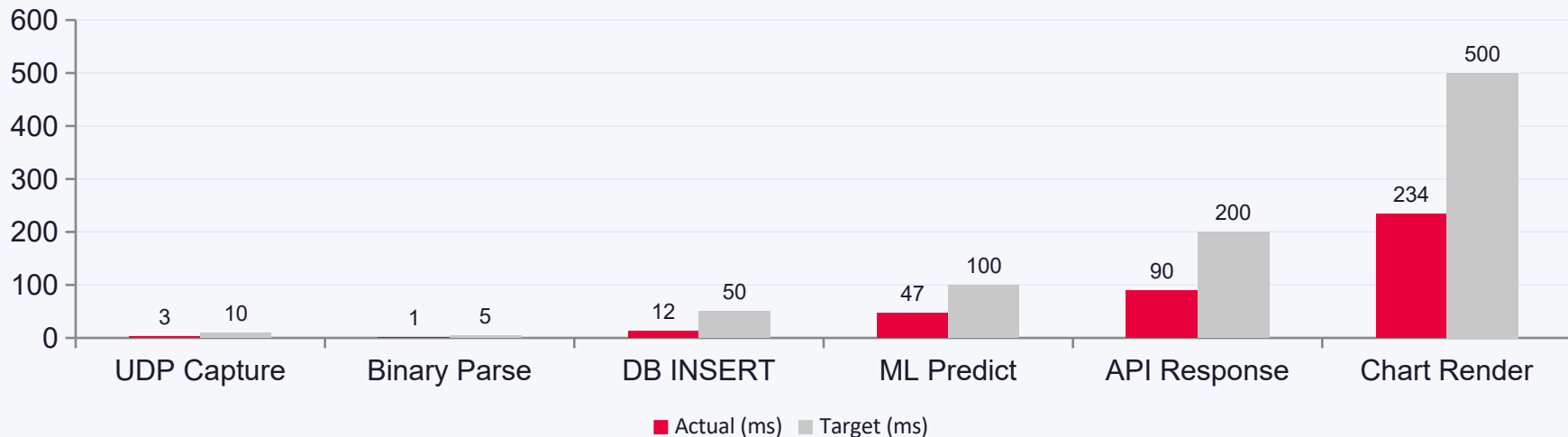
Target  $< 100ms$  — 53% faster

**98%**

**Storage Reduction**

36K  $\rightarrow$  600 samples/session

System Latency: Actual vs Target (ms) — Lower is Better



# CONCLUSION & FUTURE SCOPE

## Key Achievements

- $R^2 = 0.908$  — exceeds published research (Heilmeyer 0.89) with smaller dataset
- MAE = 1.40s — 30% better than 2.0s target; Lower than Bhowick ARIMA (2.80s)
- 98.3% storage reduction — 706 MB raw → 2.1 MB MySQL
- Zero packet loss across 548,000+ UDP packets received
- \$0 software cost vs \$200,000+ for McLaren ATLAS



## Future Roadmap

### Data Expansion

FastF1 integration → 15K+ laps; wet weather data

### LSTM Upgrade

Sequential 5-lap window; target  $R^2 > 0.95$

### Production

Docker + AWS + SSL; multi-user authentication

### Expansion

Karting, iRacing, industrial IoT domains